Group-Aware Robot Navigation in Crowds Using Spatio-Temporal Graph Attention Network With Deep Reinforcement Learning

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***Abstract*—Robots are becoming essential in human environments, requiring them to behave in a socially compliant manner. Although previous learning-based methods have shown potential in social navigation, most have treated pedestrians as individuals, failing to account for group level interactions. Additionally, these methods have modeled pairwise interactions only in the spatial domain, overlooking the temporal evolution of relations among agents. In this letter, the above limitations are addressed by proposing a novel spatio-temporal graph attention network that explicitly models group level interactions in both spatial and temporal domains. Specifically, a novel group-awareness mechanism is designed to model group-aware behaviors, and a new network is proposed to capture spatio-temporal features of relations among agents while leveraging the model-free deep reinforcement learning to optimize the group-aware navigation policy. The test results show that our approach outperforms the baselines in all metrics in both simulation and real-world experiments. Furthermore, quantitative analysis of questionnaire responses further verifies the benefits of our method in group awareness and social compliance.**

***Index Terms*—Robot navigation, group awareness, spatio-temporal graph attention network, deep reinforcement learning.**

I. Introduction

With advancements in robot technology, mobile robots are more being deployed in human environments [1], [2]. When navigating in crowds, robots must move socially.

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Socially aware navigation methods are divided into conventional and learning-based approaches. In conventional methods, like social force model (SFM), additional parameters are introduced to establish social comfort criteria [3]. Alternatively, learning-based methods exhibit better performance. Imitation learning (IL) and inverse reinforcement learning (IRL) teach social navigation strategies from demonstrations [4], [5], while deep reinforcement learning (DRL) seek to optimize the control policy by interacting with the environment [6]. Although major progress has been achieved in DRL-based methods, they often treat pedestrians as individuals. Group level interactions (e.g., group–robot interaction, GRI) has not been fully investigated. Studies show that up to 70% of pedestrians walk in groups [7]. Failing to consider groups can lead to inefficient navigation, such as difficulty in following or overtaking groups, as well as socially inappropriate behaviors like intruding into groups.

Furthermore, most recent learning-based methods encode interactions and uni-actions (unidirectional influence of an obstacle) only in the spatial domain, which is insufficient for social robot navigation [8], [9]. Temporal relation reasoning is crucial for robots to make correct decisions, as neglecting it can lead to unsafe and shortsighted behaviors. Recently, Liu et al. [10] designed a decentralized structural recurrent neural network (RNN) to capture temporal interactions. However, RNNs are expensive to train and need additional structures to handle spatial information. To date, reasoning about spatio-temporal relations for social robot navigation remains a challenging and under-explored area.

We propose a novel method, group-aware robot navigation (GARN), to resolve the above challenges by leveraging group level interactions to develop the robot's group-aware capability, as shown in Fig. 1. Specifically, we first detect groups, represent their spaces, and design a new reward function to encourage group-aware behaviors (e.g., cooperative passing, following, and overtaking) and to prohibit group intrusion. We then construct a new graph network that captures spatial relations between agents using graph edges with attention weights obtained from supervised learning and encodes the temporal dependence of spatial relations with long short-term memory (LSTM). Finally, the network is trained using the model-free DRL to learn group-aware navigation policies.

The main contributions are as follows: (1) A novel group-awareness mechanism (GAM) is designed, including group detection, space representation, and an original group-related reward function. (2) A new spatio-temporal graph attention network (STGAN) is proposed, enabling the robot to reason about spatio-temporal interactions and uni-actions among all

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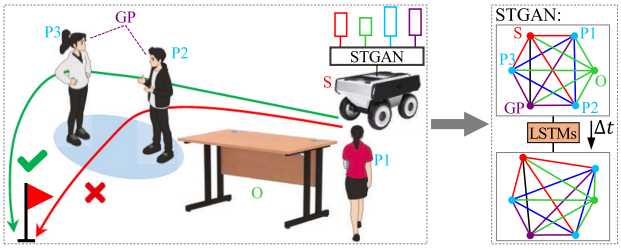

Fig. 1. An example of group-aware robot navigation using our GARN. The agents are represented as follows: robot as S, obstacle as O, individual as P_i, and group as GP. The ellipse denotes the social space of a group. All agents and their spatial relations are encoded as nodes and edges in a graph structure, STGAN. The temporal dependence of graph across different time steps is captured by LSTMs. The GARN effectively guides the robot to navigate towards its goal while avoiding collision and group intrusion.

agents while leveraging model-free DRL to optimize the navigation policy. (3) It is demonstrated that the performance of GARN in navigation efficiency, group-awareness, and social compliance exceeds state-of-the-art baselines in both simulation and real-world experiments.

The next section provides an overview of related works. Section III details our approach. In Section IV, we present evaluation experiments for our method. Discussion and conclusion are given in Sections V and VI, respectively.

II. Related Work

A. Social Robot Navigation

Previous studies proposed various methods to improve the collision avoidance ability of robot navigation [11], [12], [13]. To enhance social awareness, Truong et al. [3] proposed a social reactive control, based on SFM, for robots to navigate among individuals and groups. However, SFM-based methods suffer from tedious parameter selection and limited generalization. Conversely, IL-based approaches aim to learn control policies from expert experience [4]. Additionally, IRL was employed to model cooperative behaviors [5]. But, the effectiveness of IL and IRL-based methods relies on the quality of datasets.

Recently, DRL has gained attention due to its robustness to unpredictable events. SA-CADRL [14] is a pioneering work that employed DRL for socially aware navigation. A subsequent extension, SARL [15], designed a socially attentive module to assign relative importance to each agent and then to make control decisions according to the attention scores. Although these methods have achieved promising results, they assume the state transitions of pedestrians are known, which are, in fact, stochastic and difficult to model. In contrast, our work trains the policy network with model-free DRL, which does not require knowledge of state transitions.

Another limitation of prior studies is they mostly treat pedestrians as individuals. To address this, Vega-Magro et al. [16] employed a density function to cluster individuals into groups and adapted a navigation architecture to prevent the robot from entering groups. Nevertheless, this method is limited to static environments. Ngo et al. [17] introduced a navigation framework enabling a robot to approach both static and dynamic individuals and groups. Group-aware policies have also been proposed based on DRL [18], [19], [20]. However, these approaches only trained the robot to avoid intruding into groups and have not enhanced its ability to interact effectively with groups. Unlike [20] and other related works, this study designs an original group-related reward function to explicitly model group-robot interactions, such as cooperative passing, following, and overtaking.

B. Graph Representation Learning

Graph convolutional networks (GCNs) have received wide attention for graph representation learning. In robot navigation, using GCNs is a natural way since features of agents can be represented by nodes, while relations between agents can be encoded by edges. Chen et al. [21] proposed a GCN to enhance the representation of crowd-robot interactions based on DRL. However, their studies overlooked the temporal features of interactions. To overcome this, GCNs have been extended to spatio-temporal GCNs, which can capture both spatial and temporal relations. Wang et al. [22] proposed a spatio-temporal graph transformer for socially aware navigation. Ma et al. [23] presented a navigation architecture with a spatio-temporal reasoning module to capture spatial interactions and their temporal evolutions. Although these methods included temporal relation reasoning, they overlooked obstacle uni-actions and group level interactions. We propose STGAN to comprehensively encode obstacle uni-actions, pairwise interactions and group level interactions in both spatial and temporal domains. To the best of our knowledge, this work is the first to combine spatio-temporal GCNs with group-aware capability for social robot navigation.

III. Approach

A. Problem Formulation

1) Preliminaries: Robot navigation can be framed as a decision-making problem in a DRL framework, where a robot navigates toward a goal in an environment containing K obstacles, I pedestrians (including group members), and M groups. All agents' states are obtained through sensors and transformed into the robot-centric coordinates, with the x-axis originating at the robot's current position and extending towards its goal. Denote the robot's position as $\boldsymbol{p}_t = [p_x, p_y]$, orientation as θ_t, velocity as $\boldsymbol{v}_t = [v_x, v_y]$, radius as r, preferred speed as v_{pref}, and goal position as d^t_{gl}. The robot's state can be expressed as $\boldsymbol{S}_t = [\boldsymbol{p}_t, \theta_t, \boldsymbol{v}_t, r, v_{\text{pref}}, d^t_{\text{gl}}]$.

Two basic types of obstacles, circular and rectangular, are considered. The state of the k-th circular or rectangular obstacle is denoted as $\boldsymbol{O}^k_t = [\boldsymbol{p}^k_t, r^k_t]$ or $\boldsymbol{O}^k_t = [\boldsymbol{p}^k_t, \theta^k_t, w^k_t, l^k_t]$, where $\boldsymbol{p}^k_t = [p^k_x, p^k_y]$ represents the position, r^k_t signifies the radius, θ^k_t denotes the orientation, w^k_t and l^k_t are the width and length, respectively. More complex obstacles can be decomposed into combinations of these two basic obstacles.

Each pedestrian's state is similar to the robot's, except replacing r with personal space, lacking v_{pref} and d^t_{gl}, and adding a group ID. As shown in Fig. 2(a), unlike the traditional circular representation, personal space $\mathbb{P}^i$, is defined as an adaptive area based on asymmetric Gaussian model [24]. The group ID is denoted as $gp^i_t \in \{0, 1, \ldots, m, \ldots, M\}$, where 0 represents an individual and m implies the m-th group. We define the i-th pedestrian's state as $\boldsymbol{P}^i_t = [\boldsymbol{p}^i_t, \theta^i_t, \boldsymbol{v}^i_t, \mathbb{P}^i, gp^i_t]$.

2) Social Group Space: As shown in Fig. 2(b), static groups can be detected using the F-formation method [25]. Conversely, as depicted in Fig. 2(c), pedestrians in a dynamic group have similar motion features, which can be identified with the DBSCAN algorithm [26]. The m-th group is defined as $\mathcal{GP}^m_t =$

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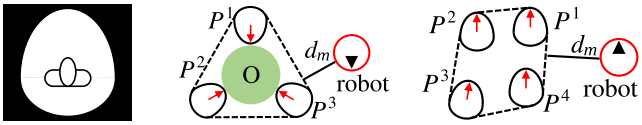

(a) Personal space (b) Static group space (c) Dynamic group space

Fig. 2. Social spaces of individual and groups. The dashed lines represent group space, delineated by a convex hull of personal spaces, and d_m is the shortest distance between the robot and the convex hull. (a) Personal space is defined as an asymmetric Gaussian ellipse. (b) Static group members face inward toward a shared o-space with close proximity. (c) Dynamic group members exhibit similar walking speeds and coincident orientations.

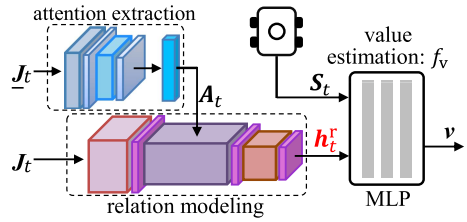

Fig. 3. The STGAN contains an attention extraction module to generate attention weights for each node, a relation modeling module to capture spatio-temporal relations, and a value estimation module to select action.

$\{i \in \{1,\ldots,I\} \mid gp_t^i = m\}$, where gp_t^i is a function that indicates each pedestrian's group ID. Given $\mathbb{P}^i$ of each member in the group, the group space $\mathbb{GP}_t^m$, is extracted as:

$$\mathbb{GP}_t^m = \text{Convexhull}(\{\mathbb{P}^i \mid i \in \mathcal{GP}_t^m\}). \tag{1}$$

In this study, $\mathbb{GP}_t^m$ is represented by a vector containing a set of points' positions. The group's state is denoted as $\boldsymbol{GP}_t^m = [\boldsymbol{p}_t^m, \boldsymbol{v}_t^m, \mathbb{GP}_t^m, n_t^m]$, where $\boldsymbol{p}_t^m$ is the gravity center of the convex hull, $\boldsymbol{v}_t^m$ denotes the mean velocity of group members, and n_t^m represents the number of group members.

3) Model-Free DRL: The joint states of $1+K+I+M$ agents are denoted as $\boldsymbol{J}_t = [\boldsymbol{S}_t, \ldots, \boldsymbol{O}_t^k, \ldots, \boldsymbol{P}_t^i, \ldots, \boldsymbol{GP}_t^m, \ldots]$. In each episode, the robot begins from an initial state and takes an action, $\boldsymbol{a}_t$, according to the learned navigation policy, $\boldsymbol{v}_t = \boldsymbol{a}_t = \boldsymbol{\pi}(\boldsymbol{J}_t)$. After a time interval, Δt, the robot transitions to the next state, $\boldsymbol{J}_{t+\Delta t}$, and receives a reward, $R(\boldsymbol{J}_t, \boldsymbol{a}_t)$. The optimal policy, $\boldsymbol{\pi}^* : \boldsymbol{J}_t \mapsto \boldsymbol{a}_t$, is formulated as follows:

$$\boldsymbol{\pi}^*(\boldsymbol{J}_t) = \underset{\boldsymbol{a}_t \in \boldsymbol{ACT}}{\arg\max} \Big[R(\boldsymbol{J}_t, \boldsymbol{a}_t) + \gamma^{\Delta t \cdot v_{\text{pref}}} \max_{\boldsymbol{a}_{t+\Delta t}} \boldsymbol{Q}^*(\boldsymbol{J}_{t+\Delta t}, \boldsymbol{a}_{t+\Delta t}) \Big], \tag{2}$$

where $\boldsymbol{ACT}$ is the action space, γ is a discount factor, and $\boldsymbol{Q}^*$ is the optimal action value function. Double deep Q-learning, a model-free policy gradient algorithm, is adopted for action value function estimation and policy learning [27].

B. Reward Function Reshaping

The robot's primary objective is to reach its goal safely, socially, and efficiently. As a result, the reward function must be reshaped to ensure a positive reward for successful goal attainment while penalizing collisions and group intrusions, and promoting socially aware and highly efficient behaviors. $R(\boldsymbol{J}_t, \boldsymbol{a}_t)$ is defined by incorporating four components:

$$R(\boldsymbol{J}_t, \boldsymbol{a}_t) = R_{\text{gl}} + R_{\text{obs}} + R_{\text{prox}} + R_{\text{grp}}. \tag{3}$$

where R_{gl} is the goal reward; R_{obs} is the obstacle reward; and R_{prox} is the proximity reward. The details of R_{gl} R_{obs}, and R_{prox} can be referred to our previous works [8], [20], [28]. The novel group-related reward R_{grp}, which consists of three terms, is proposed to encourage avoidance of group intrusion ($R_{\text{grp}}^{\text{intru}}$), overtaking or following a group ($R_{\text{grp}}^{\text{ot/flw}}$), and cooperative passing ($R_{\text{grp}}^{\text{cops}}$). R_{grp} is formulated as:

$$R_{\text{grp}} = R_{\text{grp}}^{\text{intru}} + R_{\text{grp}}^{\text{ot/flw}} + R_{\text{grp}}^{\text{cops}}, \tag{4}$$

$$R_{\text{grp}}^{\text{intru}} = \sum\nolimits_{\forall m \in M} \{-0.25 * 1_{d_m}\}, \tag{5}$$

where d_m represents the shortest distance from the robot to a convex hull, as shown in Fig. 2, and 1_{d_m} is a binary indicator function to check whether or not the group intrusion occurs.

$R_{\text{grp}}^{\text{ot/flw}}$, which incentivizes overtaking without collision and intrusion while penalizing distant following, is given by:

$$R_{\text{grp}}^{\text{ot/flw}} = \sum_{\forall m \in M_1} 0_{d_m} c_1 \rho^m \frac{\Delta(\boldsymbol{p}_{t-1}, \boldsymbol{p}_{t-1}^m) - \Delta(\boldsymbol{p}_t, \boldsymbol{p}_t^m)}{v_{\text{pref}}} \cdot \frac{\boldsymbol{v}_t}{\|\boldsymbol{v}_t\|}, \tag{6}$$

$$\rho^m = \begin{cases} 1 & \text{if } \|\Delta(\boldsymbol{p}_t, \boldsymbol{p}_t^m)\| \le d_{\text{frt}} \\ 0 & \text{otherwise} \end{cases}, \tag{7}$$

where $\Delta(\boldsymbol{p}_t, \boldsymbol{p}_t^m) = \boldsymbol{p}_t - \boldsymbol{p}_t^m$, M_1 represents a set of groups walking in the same direction as the robot while being positioned in front of it, 0_{d_m} denotes the logical negation of 1_{d_m}, c_1 is a hyperparameter for weighting lagging behind and successful overtaking, and the dot product with $\boldsymbol{v}_t$ projects the displacement difference onto the direction of the robot's velocity. Here, ρ^m is a binary function, and d_{frt} denotes the consideration range for surrounding groups.

$R_{\text{grp}}^{\text{cops}}$, which encourages cooperative passing with a group rather than waiting when encountering it, is defined as:

$$R_{\text{grp}}^{\text{cops}} = \sum_{\forall m \in M_2} 0_{d_m} c_2 \rho^m \frac{\Delta(\boldsymbol{p}_t, \boldsymbol{p}_t^m) - \Delta(\boldsymbol{p}_{t-1}, \boldsymbol{p}_{t-1}^m)}{v_{\text{pref}}} \cdot \frac{\boldsymbol{v}_t^\top}{\|\boldsymbol{v}_t\|}, \tag{8}$$

where M_2 represents a set of groups that approaches the robot from its front, and c_2 is a hyperparameter for weighting the cooperative passing penalty.

C. Network Architecture

The STGAN, containing attention extraction, relation modeling, and value estimation modules, is shown in Fig. 3. The structure of STGAN is entirely novel and fundamentally distinct from our previous work [20] in several aspects. First, while [20] employs a multilayer perceptron (MLP) architecture, this study utilizes GCNs, which are more efficient for embedding agents and their interactions. Second, this work incorporates temporal relation reasoning for enhanced spatio-temporal interaction modeling, whereas [20] considers only spatial interactions. Third, the proposed network is trained with model-free DRL, different with the model-based DRL in [20]. In this subsection, we first introduce the use of spatio-temporal GCN for reasoning relations between agents, and then provide a detailed explanation for each module.

1) Spatio-Temporal GCN: The spatio-temporal GCN is designed to model the robot navigation scenario by encoding interactions and uni-actions among all agents, as shown in Fig. 4(a). The graph is denoted as $(\boldsymbol{\mathcal{V}}_t, \boldsymbol{\mathcal{E}}_t)$, where the nodes $\boldsymbol{\mathcal{V}}_t$ correspond to the latent state representations of all agents, with $|\boldsymbol{\mathcal{V}}_t| = N = 1+K+I+M$. The edges $\boldsymbol{\mathcal{E}}_t$ capture spatial relations at the same time step, such as $\varepsilon_{t(i,j)} \in \boldsymbol{\mathcal{E}}_t$, indicating the attention the i-th agent pays to the j-th agent.

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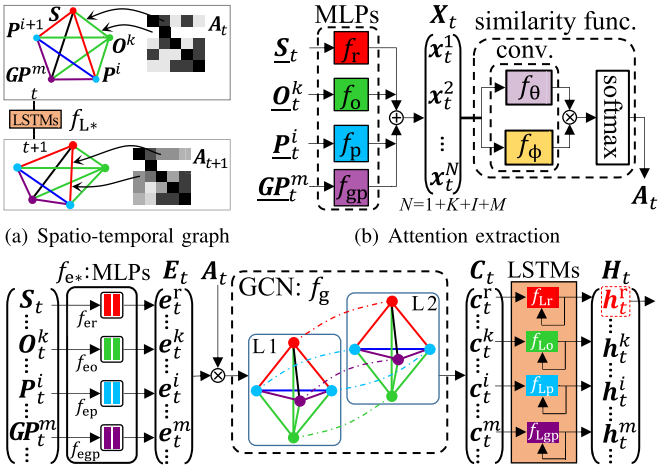

(a) Spatio-temporal graph (b) Attention extraction

(c) Relation modeling

Fig. 4. The details of STGAN structure. (a) The agents and their spatial relations are modeled as nodes and edges in the GCN, with edges of the same type depicted using the same colored lines. The strength of spatial relations is aggregated in $\boldsymbol{A}_t$, where darker colors indicate stronger connections. Temporal functions $f_{\mathrm{L}*}$ connect GCNs across different time steps. (b) The attention matrix $\boldsymbol{A}_t$, defining weights for aggregating features across different agents, is derived with a similarity function and then applied to the GCN. (c) Spatio-temporal relations between agents are inferred by concatenating through MLPs, GCN, and LSTMs networks.

The robot, individuals, and groups influence each other in various ways, so the edges of GRI, human–robot, human–human, human–group, and group–group interactions are encoded distinctly. In contrast, obstacles exert unidirectional effects on other agents equally and are not influenced by them. Thus, the obstacle–agent uni-actions are encoded uniformly. To reduce the number of trainable parameters, edges of the same type, depicted with the same colored lines in Fig. 4(a), share common function parameters. The values of spatial relations (edges) are inferred with the attention extraction module and collected in the attention matrix $\boldsymbol{A}_t$.

As the robot and pedestrians move, $(\boldsymbol{\mathcal{V}}_t, \boldsymbol{\mathcal{E}}_t)$ evolves over time. To capture the temporal dependence of spatial graphs across time steps, the LSTM networks $f_{\mathrm{L}*}$ are employed. The $f_{\mathrm{L}*}$ store relevant past information within their hidden states and discard less significant details, enabling the robot to make decisions with foresight. This approach overcomes the limitations of previous methods, which only encode spatial relations and therefore lack a long-term perspective.

2) Attention Extraction Network: The attention matrix $\boldsymbol{A}_t$, which reflects the strength of spatial relations between nodes (agents), is obtained by the attention extraction module, as shown in Fig. 4(b). The input feature vectors for this subnetwork are partial states of agents, denoted as $\underline{\boldsymbol{S}}_t = [\boldsymbol{p}_t, \theta_t, \boldsymbol{v}_t]$, $\underline{\boldsymbol{O}}_t^k = [\boldsymbol{p}^k, \theta^k]$, $\underline{\boldsymbol{P}}_t^i = [\boldsymbol{p}_t^i, \theta_t^i, \boldsymbol{v}_t^i]$, and $\underline{\boldsymbol{GP}}_t^m = [\boldsymbol{p}_t^m, \boldsymbol{v}_t^m]$. Since the dimensions of these vectors are different, four MLPs, f_{r}, f_{o}, f_{p}, and f_{gp}, are applied to generate output feature vectors with a fixed length, which are then aggregated to form the attention feature matrix $\boldsymbol{x}_t$. We then apply an embedded Gaussian as the similarity function f to extract the attention matrix [29]. The spatial attention is calculated as follows:

$$f(\boldsymbol{x}^i, \boldsymbol{x}^j) = e^{f_\theta(\boldsymbol{x}^i)^{\mathrm{T}} f_\phi(\boldsymbol{x}^j)}, \tag{9}$$

where $\boldsymbol{x}^i = \boldsymbol{x}_t[i,:]$, $f_\theta(\boldsymbol{x}^i) = \boldsymbol{W}_\theta \boldsymbol{x}^i$ and $f_\phi(\boldsymbol{x}^j) = \boldsymbol{W}_\phi \boldsymbol{x}^j$ are two convolutional embedding functions. In the matrix form, $\boldsymbol{A}_t = \mathrm{softmax}(\boldsymbol{x}_t^{\mathrm{T}} \boldsymbol{W} \boldsymbol{x}_t)$. Here, $\boldsymbol{W} = \boldsymbol{W}_\theta^{\mathrm{T}} \boldsymbol{W}_\phi$ is the trainable weight matrix. It is important to note that the size of $\boldsymbol{W}$ is only determined by the length of output feature vectors of the four MLPs and independent of the total number of agents, N. Thus, the attention extraction subnetwork can easily scale as N increases, simply by expanding the row size of $\boldsymbol{x}_t$, without the need for re-training this subnetwork.

3) Relation Modeling: The spatio-temporal interactions and uni-action among all agents are encoded through the relation modeling module, as shown in Fig. 4(c). The state of each agent is first passed through an MLP function $f_{\mathrm{e}*}$ to embed it into a latent space with a fixed length. The obtained latent matrix $\boldsymbol{E}_t$ is then multiplied by $\boldsymbol{A}_t$ and fed into the GCN f_{g} to propagate information across nodes and produce the spatial relation feature matrix $\boldsymbol{C}_t$. According to the layer-wise update rule, the output of each layer is expressed as:

$$\boldsymbol{C}_t^{(l+1)} = \sigma(\boldsymbol{A}_t \boldsymbol{C}_t^{(l)} \boldsymbol{W}_g^{(l)}), \tag{10}$$

where $\sigma(\cdot)$ is the ReLU activation function, $\boldsymbol{C}_t^{(l)}$ is the input of the l-th layer (let $\boldsymbol{C}_t^{(1)} = \boldsymbol{E}_t$), and $\boldsymbol{W}_g^{(l)}$ is a weight matrix. Finally, each part of $\boldsymbol{C}_t$ is fed into an LSTM, a temporal function $f_{\mathrm{L}*}$, to generate the spatio-temporal relation feature matrix $\boldsymbol{H}_t$, which is a collection of LSTMs' hidden states.

Note that the size of weight matrix $\boldsymbol{W}_g^{(l)}$ is determined by the length of input and output feature vectors of nodes in f_{g} and does not change with the number of nodes. Thus, handling with a variable number of agents is straightforward in our approach by adjusting the number of nodes in f_{g}.

4) Value Estimation: The output relation feature vector in the first row of $\boldsymbol{H}_t$, $\boldsymbol{h}_t^{\mathrm{r}} = \boldsymbol{H}_t[0,:]$, represents the aggregated influence of obstacles, individuals, and groups on the robot in both spatial and temporal domains. It is concatenated with the robot state $\boldsymbol{S}_t$ before being fed into the value estimation module f_{v}, which is an MLP embedding function as shown in Fig. 3. The output predicts the value of the robot state and determines the action command $\boldsymbol{v}_t$ within an action space.

IV. Experiments

In this section, the performance of the proposed approach is evaluated in both simulation and real-world experiments.

A. Experiment Setup

1) Simulation Environment: The robo-gym simulation environment [30], which integrates ROS, Gazebo, and OpenAI Gym, was used to train and evaluate the proposed approach. This platform provides simulating scenarios where a robot navigates to reach its goal, and supports sensor plugins, such as 3D LiDAR and depth camera, offering a realistic data perception system. This platform was further extended by implementing a custom Python version of the extended SFM to simulate human groups. The initial and final positions (if any) of each obstacle, pedestrian, and group were randomly placed within a 10-meter square space. The robot's r, v_{pref}, Δt, and t_{limit} were set to 0.25 m, 1 m/s, 0.25 s, and 25 s, with start and goal positions at $(0, 0.5)$ and $(2, 9.5)$, respectively.

2) Implementation Details: The hyperparameters of c_1 and c_2 were set to 0.05 and 0.025, respectively. The sizes of hidden layers in f_{r}, f_{o}, f_{p}, f_{gp}, f_{er}, f_{eo}, f_{ep}, f_{egp}, and f_{v} were designated as (256, 128), (128, 128), (256, 128), (128, 128), (512, 256), (256, 256), (512, 256), (512, 256), and (256, 128, 128), separately. Both $\boldsymbol{W}_\theta$ and $\boldsymbol{W}_\phi$ had dimensions of 128.

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TABLE I
QUANTITATIVE RESULTS OF TESTS IN SIMULATION

	Methods	$r_s\uparrow$	$r_c\downarrow$	t_s(s)$\downarrow$	d_o(m)$\uparrow$	$n_{dis}\downarrow$	$n_i\downarrow$	$n_{cp}\uparrow$	$n_{flw}\uparrow$	$n_{ot}\uparrow$
S1	SOADRL	0.88	0.10	**12.01**	0.18	41/500	25/500	1/500	1/500	2/500
	GAP	0.92	0.07	15.53	**0.26**	11/500	**0**/500	4/500	2/500	4/500
	FSTRL	0.95	0.04	15.27	0.25	28/500	23/500	2/500	1/500	3/500
	GARN*	**1.00**	**0.00**	12.87	0.22	**2**/500	**0**/500	**29**/500	**8**/500	**25**/500
S2	SOADRL	0.53	0.45	23.42	0.02	98/500	73/500	2/500	3/500	0/500
	GAP	0.72	0.27	24.08	**0.24**	25/500	12/500	7/500	6/500	2/500
	FSTRL	0.78	0.15	23.96	**0.24**	69/500	63/500	3/500	4/500	1/500
	GARN*	**0.97**	**0.00**	**20.83**	0.21	**7**/500	**0**/500	**72**/500	**23**/500	**19**/500

Notes: (*) refers to our method with full modules. The S1 scenario contains 3 individuals, 6 obstacles, 1 static group (2 members), and 1 dynamic group (3 members), while the S2 scenario consists of 6 individuals, 8 obstacles, 2 static groups (2 and 3 members), and 2 dynamic groups (3 and 4 members).

TABLE II
TEST RESULTS OF ABLATION STUDIES FOR NETWORK

Methods	SN1					SN2				
	$r_s\uparrow$	$r_c\downarrow$	t_s(s)$\downarrow$	$n_{dis}\downarrow$	$n_i\downarrow$	$r_s\uparrow$	$r_c\downarrow$	t_s(s)$\downarrow$	$n_{dis}\downarrow$	$n_i\downarrow$
A2DRL	0.92	0.02	16.98	41/500	3/500	0.79	0.06	24.81	72/500	7/500
DSRNN	0.96	0.03	14.39	36/500	48/500	0.86	0.12	22.12	65/500	92/500
STGAN*	**0.99**	**0.00**	**13.01**	**3**/500	**2**/500	**0.97**	**0.01**	**20.31**	**7**/500	**5**/500

Notes: (*) indicates our method using the STGAN without $R_{grp}^{ot/flw}$ and R_{grp}^{cops}. The SN1 scenario contains 6 individuals, 2 static groups (2 and 3 members), and 2 dynamic groups (3 and 4 members), while the SN2 scenario consists of 12 individuals, 4 static groups (2, 2, 3, and 4 members), and 4 dynamic groups (2, 3, 3, and 4 members).

The GCN f_g included two layers with output dimensions of nodes in both L1 and L2 layers set to 256 and 126, respectively. Further details on the hyperparameter determination process can be found in Section 4.2 of [28]. To stabilize training, the attention extraction subnetwork was pre-trained using supervised learning with human gaze data serving as the ground truth for attention weights [31]. Then, the relation modeling and value estimation subnetworks were jointly trained using model-free DRL, freezing the attention extraction parameters. The Adam optimizer was employed with a learning rate of 0.0005 and a discount factor γ of 0.9. Our approach and all baselines were trained over $20k$ episodes and tested with 500 episodes.

3) Metrics: A set of metrics was adopted to evaluate the performance, including the success rate (r_s) and collision rate (r_c). For successful episodes, the average navigation time (t_s), average distance from the robot to its closest obstacle (d_o), and average number of individual discomfort instances (n_{dis}) were counted. Additionally, we record the total number of instances where the robot intrudes (n_i), cooperatively passes (n_{cp}), follows (n_{flw}), and overtakes (n_{ot}) a group.

B. Simulation Experiments

1) Quantitative Evaluation: The GARN was compared with three baselines: SOADRL [32], which integrates DRL with obstacle occupancy grid maps; GAP [19], which learns group-aware policies using DRL, and FSTRL [23], which trains a spatio-temporal graph with model-free DRL. For fair comparison, obstacles were fed into FSTRL and GAP as static pedestrians of the same size, while groups, controlled by the same policy as ours, were added into SOADRL and FSTRL. All methods were trained following their original configurations in the S1 scenario and tested in S1 and S2.

The quantitative results of tests are shown in Table I. Among the methods, SOADRL behaves most aggressively, intruding into groups frequently, which results in the shortest t_s. However, its aggressiveness also leads to the lowest r_s and highest r_c. Furthermore, as the scenario complexity increases, a notable decline in SOADRL's performance is observed, likely due to its reliance on occupancy grid maps for obstacle encoding, which lacks generality. GAP and FSTRL treat obstacles and humans as individuals without distinguishing between them, leading to the overly conservative policies. As a result, both GAP and FSTRL exhibit longer t_s and larger d_o than the other two methods. In terms of r_s and r_c, FSTRL is competitive to GARN in S1 and outperforms GAP in both scenarios. This is because FSTRL and GARN utilize graph networks to encode relations in both spatial and temporal domains, whereas GAP lacks temporal representation of these interactions. However, in the more complex scenario of S2, FSTRL underperforms our method in nearly all metrics. The reason is that FSTRL only captures agents' positions without incorporating dynamic information, causing suboptimal action selection. Additionally, SOADRL and FSTRL perform poorly in all group-related metrics compared to GAP and GARN, as they do not consider groups. Frequent group intrusions in SOADRL and FSTRL also result in significant discomfort to humans, as reflected by the high n_{dis}. The results demonstrate that incorporating the GAM with STGAN greatly enhances the robot's navigation efficiency, social compliance, and group awareness.

2) Ablation Study for Network: To assess the benefits of the proposed network, we compared our ablation model STGAN—omitting the novel group-related rewards $R_{grp}^{ot/flw}$ and R_{grp}^{cops} from GARN—with two other methods: A2DRL [20], which uses a spatial attention module to reason about relations, and DSRNN [10], which models human–robot interactions by a spatio-temporal graph. All three methods were trained in the SN1 scenario and tested in SN1 and SN2, with the results shown in Table II. A2DRL exhibits the lowest r_s and highest t_s, with performance deteriorating significantly as the number of individuals and groups increases. This is due to A2DRL's focus on spatial interactions, neglecting temporal reasoning, which results in shortsighted navigation. A2DRL and STGAN achieve similar performance in n_i, as they use the same reward functions to train avoiding group intrusion. DSRNN outperforms A2DRL in r_s and t_s by encoding both spatial and temporal interactions, but underperforms in n_i, as it does not account for groups. This limitation also leads to its highest r_c. In the sparse SN1, STGAN and DSRNN show comparable results in r_s, with STGAN slightly better in t_s. However, in the denser SN2, STGAN exhibits clear advantages over DSRNN across all metrics. This is because the structural RNNs used in DSRNN suffer from forgetting issues and difficulty in processing global-level interactions, leading to suboptimal navigation policies in dense scenarios. Besides, STGAN significantly outperforms A2DRL and DSRNN in n_{dis}, due to its newly proposed personal space modeling. This ablation study verifies the advantages of STGAN structure: (1) Compared to methods like A2DRL that model only spatial interactions, STGAN achieves greater navigation efficiency across all scenarios. (2) It also outperforms methods, such as DSRNN, which use RNNs to capture spatial and temporal interactions, particularly in complex scenarios.

3) Ablation Study for GAM: Table III presents the results of ablation studies conducted to evaluate the effectiveness of the GAM, which is implemented on the same network as SARL [15]. Our previous work, A2DRL, has also been adapted to use the SARL network to serve as a baseline. All methods are trained in

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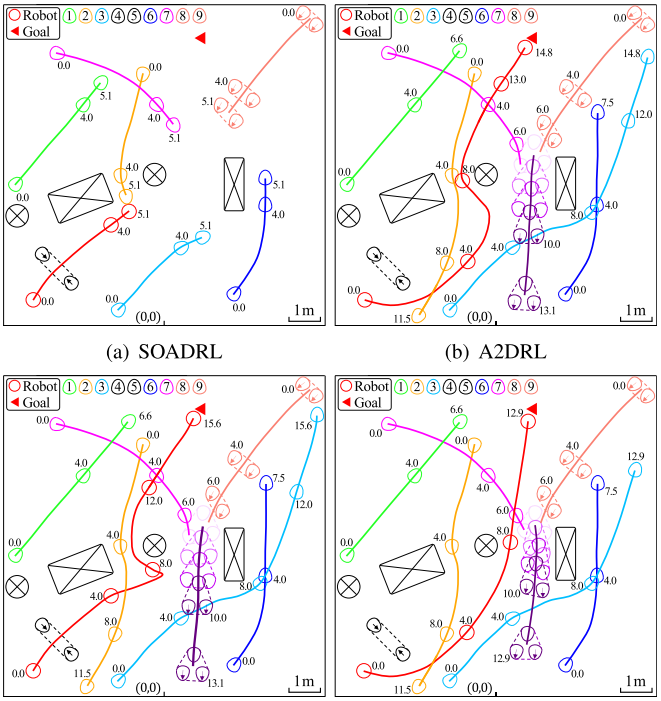
TABLE III
TEST RESULTS OF ABLATION STUDIES FOR GROUP-AWARENESS

	Methods	r_s↑	r_c↓	t_s(s)↓	n_i↓	n_{cp}↑	n_{flw}↑	n_{ot}↑
SG1	SARL	0.76	0.24	**13.64**	56/500	2/500	4/500	1/500
	A2DRL†	0.97	0.02	16.42	**0**/500	3/500	5/500	2/500
	GAM*	**1.00**	**0.00**	14.12	**0**/500	**61**/500	**14**/500	**52**/500
SG2	SARL	0.53	0.46	**21.13**	147/500	3/500	5/500	0/500
	A2DRL†	0.85	0.02	24.85	**0**/500	2/500	11/500	0/500
	GAM*	**0.98**	**0.00**	22.03	**0**/500	**129**/500	**52**/500	**39**/500

Notes: † indicates our previous A2DRL method implemented using the SARL network, while (*) represents our new method incorporating the GAM within the same network as SARL. The SG1 scenario consists of 4 individuals, 2 static groups (3 and 4 members), and 2 dynamic groups (2 and 3 members), while the SG2 scenario contains 5 individuals, 3 static groups (2, 3, and 3 members), and 4 dynamic groups (2, 3, 3, and 4 members).

the SG1 scenario and tested in SG1 and SG2. SARL performs the worst in most metrics due to its inability to account for human groups. Additionally, its frequent intrusions into groups, indicated by the highest n_i, result in the shortest t_s but also lead to the worst outcomes in r_s and r_c, especially in the denser SG2 scenario. Both A2DRL and GAM, with their reward functions reshaped to train the robot to avoid intruding into groups, achieve identical performance in n_i and outperform SARL across nearly all metrics except t_s. In SG1, A2DRL and GAM exhibit comparable results in r_s and r_c, but GAM achieves better navigation efficiency, as reflected by a shorter t_s. As the number of groups increases in SG2, GAM significantly outperforms A2DRL across most metrics except n_i. This improvement is due to GAM's novel reward function, which effectively trains the robot to cooperatively pass, follow, and overtake groups, as evidenced by its highest n_{cp}, n_{flw}, and n_{ot}. In contrast, A2DRL's focus on merely avoiding group intrusions proves insufficient for complex navigation tasks. These results underscore the importance of leveraging group level interactions to enhance the robot's group-awareness.

4) Qualitative Evaluation: The effectiveness of our approach was further evaluated through qualitative tests, where the robot's start and goal positions were randomly assigned. Sample trajectories of SOADRL, A2DRL, FSTRL, and GARN are shown in Fig. 5. The SOADRL-controlled robot intrudes into the static group to obtain a short path but collides with the first individual around 5.1 s. This collision arises from SOADRL's failure to reason about the obstacle's uni-action, which causes the individual's direction change. In contrast, A2DRL avoids intruding into the static group but lacks the capability to capture group interactions, as it has not learned to interact with groups effectively. Therefore, upon encountering a group at around 7.0 s, A2DRL fails to pass cooperatively and instead takes a detour. Similar to SOADRL, FSTRL chooses to traverse through the static group to minimize t_s; however, unlike SOADRL, it successfully avoids colliding with the first individual due to its systematic encoding of spatio-temporal relations. Despite this, FSTRL exhibits a "freezing" behavior from 6.0 s to 7.5 s and ultimately takes a detour due to its inability to interact with groups. In contrast, the GARN-controlled robot successfully navigates to the right of the dynamic group's movement direction while maintaining a comfortable distance. This, together with the group's shape reorganization process of "Λ-I-Λ ", jointly facilitates cooperative passing between 7.0 s and 9.0 s, resulting in the shortest t_s. In summary, GARN outperforms the baselines in reasoning obstacle uni-actions (vs. SOADRL), avoiding group intrusion (vs. SOADRL and FSTRL), and interacting with dynamic groups (vs. SOADRL, A2DRL and FSTRL). As a result,



(a) SOADRL (b) A2DRL (c) FSTRL (d) GARN(ours)

Fig. 5. Sample trajectories of SOADRL, A2DRL, FSTRL, and GARN in qualitative evaluation tests. Asymmetric ellipses symbolize individuals, while obstacles are depicted as rectangles and circles with crosses. Convex hulls outline groups, with members emphasized in same color. The reorganization process of the dynamic group is represented by the color gradient from light purple to dark purple. The discontinuous trajectories of individual 7 and the salmon group are due to change in group size. Individual IDs are positioned in the upper-left corner, and their locations are time-stamped.

GARN generates the most efficient, socially compliant, and group-aware paths, demonstrating the advantages of our novel GAM and STGAN.

C. Real World Experiments

To further validate the advantages of our method, real world experiments of FSTRL, A2DRL, and our GARN were conducted. In addition to metrics used in simulation, feedback from participants was collected to assess the social compliance of each method. The real-world experiments have been reviewed by Institutional Review Board of JUST.

1) Robot and Environment Setup: The experiment platform is a mobile robot, equipped with a 3D LiDAR and a RealSense D455, as shown in Fig. 6(a). We applied YOLOv5 for agent detection and used Kalman filters to estimate the positions and sizes of agents by processing the aligned LiDAR scans and depth images [33]. Pedestrians' velocities were estimated using optical flow [34] and groups were detected based on the method discussed in Section III-A2. A mobile workstation, featuring an Intel Core i9-13900 K and an RTX 3090, was used to ensure sufficient real-time computation power for network inference and pedestrian detection. The first set of experiments was conducted in a controlled environment (15 m*18 m) to validate the effectiveness of the proposed method, as shown in Fig. 6(b). To further evaluate the approach's generalizability, a second set of experiments was performed in a natural environment (25 m*30 m), a hall in a teaching building with various dynamic individuals and groups, as shown in Fig. 6(c).

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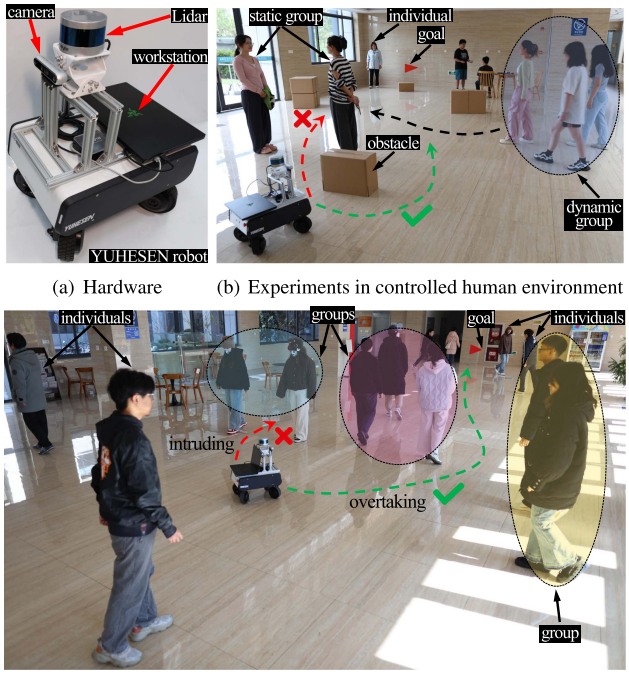

(a) Hardware (b) Experiments in controlled human environment

(c) Experiments in natural human environment

Fig. 6. Real-world experiments. (a) is the experiment hardware. (b) and (c) are experiments in the controlled and natural environments, respectively.

2) Experiment Design: For experiments in the controlled environment, we recruited 40 participants and divided them into 5 teams. Each team contained one individual, two static groups, and a dynamic group. During the experiments, the robot and each team of participants were tasked with completing point-to-point navigation exercises, with participants encouraged to behave naturally as they would in everyday life. We designed 10 experiment scenarios in which the robot, obstacles, individual and groups were assigned different start and goal positions (if any). For each scenario, five tests were conducted for FSTRL, A2DRL, and GARN, resulting in a total of 15 tests. The order in which the three methods controlled the robot was randomized for each team in every scenario, ensuring participants were unaware of which method was active. In the natural environment experiments, each method was tested for 80 episodes on 4 weekdays, when pedestrians were consistently present in the hall.

3) Evaluation: In the controlled environment experiments, participants were asked to fill a questionnaire immediately after each test, rating the robot's behavior in terms of group-awareness, naturalness and comfort [35]. The ratings were quantified using the Likert Scale [36], ranging from 1 (strongly disagree) to 7 (strongly agree). The advantages of GARN in ensuring group-aware and socially compliant robot behavior are evident, as shown in Fig. 7. It is also clear that FSTRL's frequent group intrusions result in low group-awareness and comfort ratings, while A2DRL's lack of temporal relation encoding leads to unnatural behavior.

For experiments in the natural human environment, where collecting questionnaires is impractical, we rely on metrics to compare the performance of each method. Besides the metrics defined in subsection IV-A3, we also recorded whether the robot ran a timeout (n_{to}, defined as exceeding 80 s). The results

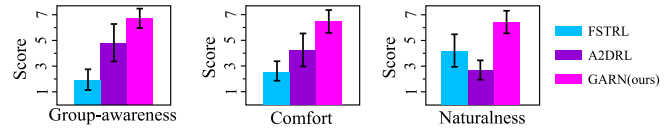

Fig. 7. Quantitative analysis of questionnaires from controlled environment experiments. Group-awareness is the ability to navigate around groups approximately without disruption. Comfort reflects the absence of annoyance experienced by participants, while naturalness refers to the similarity between the robot and humans in behavior.

TABLE IV
RESULTS OF EXPERIMENTS IN THE NATURAL ENVIRONMENT

Methods	$r_{\mathrm{s}}\uparrow$	$r_{\mathrm{c}}\downarrow$	$n_{\mathrm{to}}\downarrow$	t_{s}(s)$\downarrow$	$n_{\mathrm{i}}\downarrow$	$n_{\mathrm{cp}}\uparrow$	$n_{\mathrm{flw}}\uparrow$	$n_{\mathrm{ot}}\uparrow$
FSTRL	48/80	22/80	10/80	46.79	17/80	1/80	3/80	1/80
A2DRL	42/80	3/80	35/80	59.61	**0**/80	2/80	2/80	0/80
GARN(ours)	**73**/80	**1**/80	**6**/80	**41.52**	**0**/80	**16**/80	**7**/80	**23**/80

are shown in Table IV. FSTRL outperforms A2DRL in some navigation efficiency metrics, such as r_{s}, n_{to}, and t_{s}, due to its spatio-temporal relation reasoning, but underperforms in group-related metrics, as it lacks considering groups, which also leads to a higher r_{c}. Consistent with the simulation results, our GARN surpasses both baselines across all metrics, attributed to the proposed STGAN and GAM. Additionally, a performance gap between simulation and real-world experiments is observed for FSTRL but not for GARN, as our training system accounts for sensing noise and motion uncertainties in the simulation environment, enhancing the method's sim-to-real generalizability.

V. DISCUSSION

A. Challenges in Group Detection

Our approach relies on several assumptions, such as members of the same dynamic group share coincident orientations and similar walking speeds, for detecting groups. However, these assumptions may not always hold. A potential solution to this limitation is adopting data-driven methods, such as machine learning, to improve group identification. Additionally, mutual occlusions among pedestrians restrict our method to detecting groups with no more than four members.

B. Failure Scenarios

Although our method shows significant improvements over baseline approaches, we observed several situations where the robot failed to reach its goal in real-world experiments. The primary cause of failure was timeout, typically occurring in high-density pedestrian scenarios in which no feasible path existed. One potential solution to this issue is enabling the robot to actively create passable routes, such as requesting pedestrians to move aside or removing movable obstacles. Additionally, a collision occurred due to the sudden appearance of a pedestrian from an elevator adjacent to the robot's path, causing another failure. While no harm resulted from this incident, as the robot's bump sensor triggered an emergency stop, such incidents should be prevented in real-world applications. One possible solution is to train the robot to navigate cautiously in areas where pedestrians may appear suddenly or behave unpredictably.

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VI. Conclusion

In this letter, we present the GARN approach to enhance the robot's navigation performance, particularly in terms of group-awareness. By incorporating a novel GAM with a newly designed group-related reward function, the robot learns to exhibit group-aware behaviors and avoid group intrusions. Additionally, by constructing a new graph network, STGAN, spatio-temporal relations among agents are systematically reasoned to determine the optimal navigation policy. The effectiveness of GAM and STGAN was validated through extensive ablation studies. The GARN approach, which integrates GAM with STGAN, outperforms state-of-the-art methods in navigation efficiency, group-awareness, and social compliance, as demonstrated through both simulation and real world experiments. Finally, a quantitative analysis of participants' questionnaire responses provides an intuitive evaluation of each method's performance.

For future work, the following directions will be pursued: (1) We plan to detect groups using GCNs, enabling a fully graph-based method. A promising extension is the use of hypergraphs, where a single hyperedge can connect more than two nodes (e.g., representing a group with five members). (2) We aim to improve navigation efficiency in denser environments by developing an "active path clearing" strategy, which may include actions such as requesting passage via a speaker or utilizing a manipulator to remove obstacles.

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